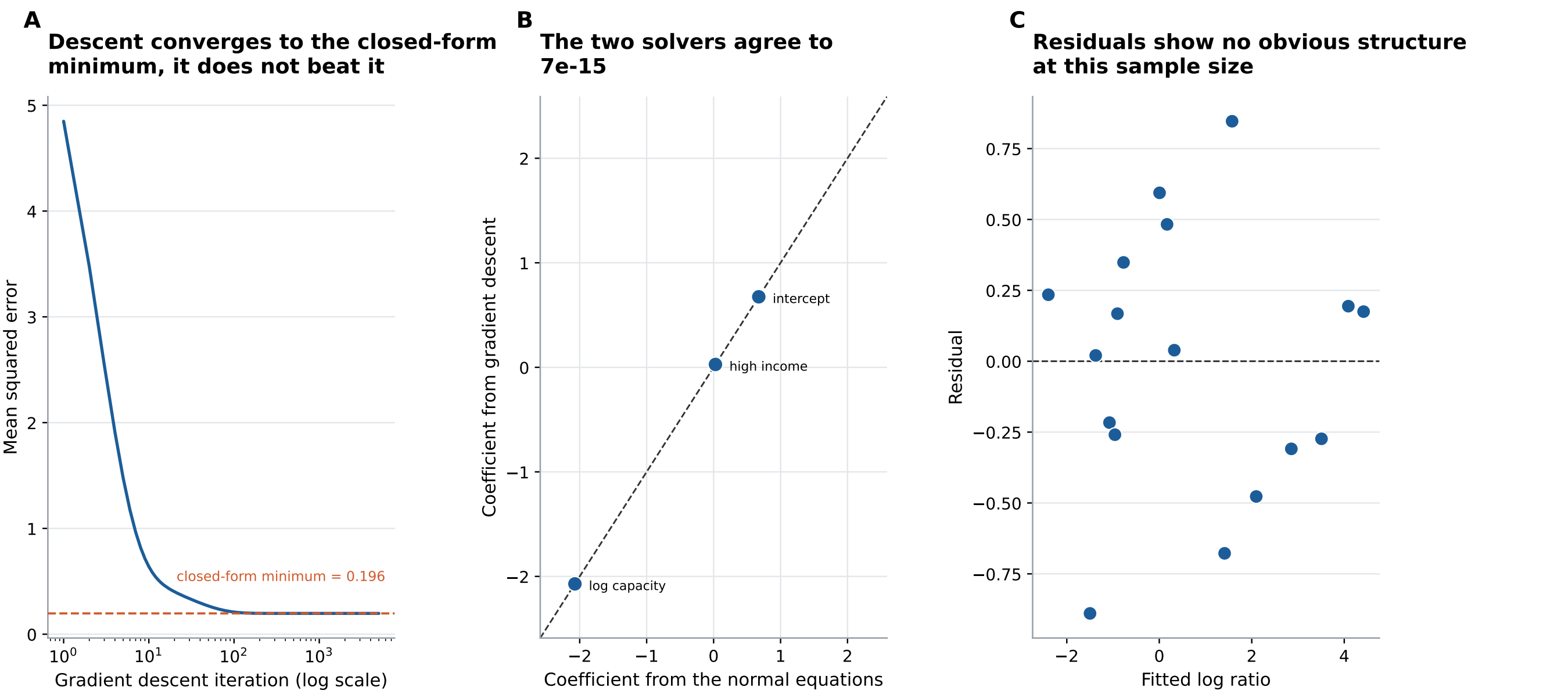


The country-level regression, solved two ways as a check



Outcome is the natural log of the burden-to-capacity share ratio for the 17 countries with both a burden and a positive capacity value. Predictors are standardized so a single learning rate suits both. Panel A: gradient descent minimizes the same mean squared error the normal equations solve exactly, so the curve approaches the dashed line from above and never crosses it. Panel B: coefficients from the two solvers plotted against each other; points on the diagonal mean they agree. Panel C: residuals against fitted values, the standard check for non-constant variance or curvature, neither of which is visible here, though 17 points is too few to see much. Capacity appears in the denominator of the outcome, so its large negative coefficient is mechanical rather than a substantive finding.